

CLOSURES OF CONJUGACY CLASSES IN CLASSICAL REAL LINEAR LIE GROUPS

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1. CLASSICAL GROUPS

By a classical group we mean a group belonging to one of the ten infinite series of complex or real Lie groups listed below. The index j ($1 \leq j \leq 10$) is used throughout to refer to one of these ten series. Each classical group G is given by its natural representation in a finite dimensional (right) F -vector space V where $F = \mathbb{R}, \mathbb{C}$, or $\mathbb{H}$.

The first three series are the general linear groups:

$$\begin{aligned} j = 1 & : GL_n(\mathbb{C}), & n \geq 0, F = \mathbb{C}; \\ j = 2 & : GL_n(\mathbb{R}), & n \geq 0, F = \mathbb{R}; \\ j = 3 & : GL_n(\mathbb{H}), & n \geq 0, F = \mathbb{H}. \end{aligned}$$

For the remaining seven series V is equipped with a non-degenerate form $f : V \times V \rightarrow F$ and G is the subgroup of the general linear group consisting of all automorphisms of V which preserve f . The form f may be symmetric, skew-symmetric, hermitian, or skew-hermitian.

The remaining seven series are:

$$\begin{aligned} j = 4 & : U(k, n-k), 0 \leq k \leq n, f = \mathbb{C}, f \text{ hermitian}; \\ j = 5 & : O_n(\mathbb{C}), n \geq 0, F = \mathbb{C}, f \text{ symmetric}; \\ j = 6 & : O(k, n-k), 0 \leq k \leq n, F = \mathbb{R}, f \text{ symmetric}; \\ j = 7 & : O^*(2n), n \geq 0, F = \mathbb{H}, f \text{ skew-hermitian}; \\ j = 8 & : Sp_{2n}(\mathbb{C}), n \geq 0, F = \mathbb{C}, f \text{ skew-symmetric}; \end{aligned}$$

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$$\begin{aligned} j &= 9 : \text{Sp}_{2n}(\mathbb{R}), \quad n \geq 0, \quad F = \mathbb{R}, \quad f \text{ skew-symmetric;} \\ j &= 10 : \text{Sp}(k, n-k), \quad 0 \leq k \leq n, \quad F = \mathbb{H}, \quad f \text{ hermitian.} \end{aligned}$$

If $j = 8$ or 9 we have $\dim_{\mathbb{F}} V = 2n$, while in all other cases $\dim_{\mathbb{F}} V = n$.

2. STATEMENT OF THE PROBLEM

Let G be a classical group and L its Lie algebra. We view L as a G -module via the adjoint representation. Denote by O an orbit of G in L . Let G act on itself via inner automorphisms and denote by C an orbit of G in G , i.e., C is a conjugacy class of G . We shall denote by $\bar{O}$ and $\bar{C}$ the closures of O and C , respectively, for the ordinary topology of L and G . When G is a complex Lie group ($j = 1, 5, 8$) then this closure is the same as the Zariski closure.

The closure $\bar{O}$ is a finite (disjoint) union of G -orbits and the closure $\bar{C}$ is a finite (disjoint) union of G -conjugacy classes. Our problem is to determine which orbits resp. conjugacy classes are contained in $\bar{O}$ resp. $\bar{C}$. If O_1 and O_2 are two orbits of G in L let us write $O_1 \leq O_2$ if $O_1 \subset \bar{O}_2$. Then clearly, $\leq$ is a partial order in the set of all G -orbits in L and our problem reduces to the problem of describing this partial order. Similarly, we define the partial order $\leq$ in the set of all conjugacy classes of G .

3. EIGENVALUES AND PRIMARY DECOMPOSITION

Recall that $x \in L$ is an F -linear operator in V which if $4 \leq j \leq 10$ satisfies

$$f(x(v_1), v_2) + f(v_1, x(v_2)) = 0$$

for all $v_1, v_2 \in V$.

Let $x \in L$. If $F = \mathbb{C}$ define $V(\zeta, x)$ for $\zeta \in \mathbb{C}$ to be the corresponding primary subspace of x , i.e.,

$$V(\zeta, x) = \text{Ker}(x - \zeta)^\infty.$$

If $F = \mathbb{R}$ or $\mathbb{H}$ then we define $V(\zeta, x)$ for $\zeta \in \mathbb{C}$ by

$$V(\zeta, x) = \text{Ker}(x^2 - 2 \text{Re}(\zeta)x + |\zeta|^2)^\infty.$$

We say that $\zeta \in \mathbb{C}$ is an eigenvalue of x if $V(\zeta, x) \neq 0$. If $j \in \{5, 8\}$ and $\zeta \neq 0$ then $V(\zeta, x)$ is totally isotropic. The same is true if $j \in \{4, 6, 7, 9, 10\}$ and $\zeta \neq -\bar{\zeta}$. To avoid this inconvenience, we introduce the subspaces $V'(\zeta, x)$ which are defined by

$$V'(\zeta, x) = \begin{cases} V(\zeta, x) & \text{for } j \in \{1, 2, 3\}, \\ V(\zeta, x) + V(-\zeta, x) & \text{for } j \in \{5, 8\}, \\ V(\zeta, x) + V(-\bar{\zeta}, x) & \text{for } j \in \{4, 6, 7, 9, 10\}. \end{cases}$$

For $4 \leq j \leq 10$ the subspaces $V'(\zeta, x)$ are non-degenerate and orthogonal to each other.

Thus we obtain a direct (orthogonal if $j \geq 4$) decomposition

$$V = \sum_{\zeta \in \Omega_j} V'(\zeta, x) \quad (3.1)$$

where

$$\begin{aligned} \Omega_1 &= \mathbb{C}, \\ \Omega_2 &= \Omega_3 = \{\zeta \in \mathbb{C} : \operatorname{Im} \zeta \geq 0\}, \\ \Omega_4 &= \{\zeta \in \mathbb{C} : \operatorname{Re} \zeta \geq 0\}, \\ \Omega_5 &= \Omega_8 = \{\zeta \in \mathbb{C} : \operatorname{Im} \zeta > 0\} \cup \{\zeta \in \mathbb{R} : \zeta \geq 0\}, \\ \Omega_6 &= \Omega_7 = \Omega_9 = \Omega_{10} = \{\zeta \in \mathbb{C} : \operatorname{Re} \zeta \geq 0, \operatorname{Im} \zeta \geq 0\}. \end{aligned}$$

We shall refer to the decomposition (3.1) as the primary decomposition with respect to x .

4. THE CENTRALIZER THEOREM

Let G be a classical group, L its Lie algebra, and $\operatorname{Ad}: G \rightarrow \operatorname{Aut}(L)$ the adjoint representation of G . For $a \in G$ and $x \in L$ we shall write $a \cdot x$ instead of $\operatorname{Ad}(a)(x)$. Similarly $G \cdot x$ stands for $\{a \cdot x : a \in G\}$, i.e., the G -orbit in L containing x .

The centralizer $C_G(x)$ of $x \in L$ in G consists of all $a \in G$ such that $a \cdot x = x$. The Lie algebra of $C_G(x)$ is

$$Z_L(x) = \{y \in L : [y, x] = 0\}.$$

Now assume that $x \in L$ is semisimple. For $\zeta \in \mathbb{C}$ let x_ζ (resp. x'_ζ) be the restriction of x to $V(\zeta, x)$ (resp. $V'(\zeta, x)$). Let $G(\zeta, x)$ be the general linear group of $V(\zeta, x)$ if $1 \leq j \leq 3$, and otherwise the group of all automorphisms of $V'(\zeta, x)$ which preserve the restriction of the form f to $V'(\zeta, x)$. (The form f is the form on V which defines G .) Finally, let $C(\zeta, x)$ be the centralizer of x'_ζ in $G(\zeta, x)$. The primary components $V'(\zeta, x)$ in (3.1) are invariant under $C_G(x)$ and consequently we have

$$C_G(x) = \prod_{\zeta \in \Omega_j} C(\zeta, x). \quad (4.1)$$

Our definition of classical groups finds its justification in the following theorem.

THEOREM 1. *Let G be a classical group, L its Lie algebra, and $x \in L$ a semisimple element. Then $C_G(x)$ is a (finite) direct product of classical groups. More precisely, (4.1) is valid and each $C(\zeta, x)$ is a classical group of k -th series where k depends only on j and ζ .*

(The values of k are given in Table I below.)

Proof. It remains to prove the claim about $C(\zeta, x)$.

If $j = 1$ or if $j \in \{2, 3\}$ and $\zeta \in \mathbb{R}$ then $C(\zeta, x)$ is the general linear group of $V(\zeta, x)$. If $j = 2$ and $\zeta \notin \mathbb{R}$ then $C(\zeta, x)$ is the general linear group of $V(\zeta, x)$ viewed as a vector space over the field $\mathbb{R}[x_\zeta]$, which is isomorphic to $\mathbb{C}$. When $j = 3$ and $\zeta \notin \mathbb{R}$ there exists a basis $e_0, e_1, \dots, e_m$ of $V(\zeta, x)$ such that $x(e_k) = e_k \zeta$ for all k . Let W be the complex linear span of $e_0, e_1, \dots, e_m$. Then $C(\zeta, x)$ is isomorphic to the general linear group of W .

If $j = 4$ and $\operatorname{Re} \zeta = 0$ then $C(\zeta, x)$ is the group of automorphisms of $V(\zeta, x)$ which preserve the restriction of f to $V(\zeta, x)$. If $j = 4$ and $\operatorname{Re} \zeta \neq 0$ then $C(\zeta, x)$ is isomorphic to the general linear group of $V(\zeta, x)$.

If $5 \leq j \leq 10$ then $C(0, x)$ is the group of automorphisms of $V(0, x)$ which preserve the restriction of f to $V(0, x)$. If $j \in \{5, 8\}$ and $\zeta \neq 0$ then $C(\zeta, x)$ is isomorphic to the general linear group of $V(\zeta, x)$.

Now let $j \in \{6, 7, 9, 10\}$. If $\zeta \in \mathbb{R}$ and $\zeta \neq 0$ then $C(\zeta, x)$ is isomorphic to the general linear group of $V(\zeta, x)$. If $\zeta \notin \mathbb{R}$ and $\operatorname{Re} \zeta \neq 0$ then $C(\zeta, x)$ is isomorphic to the centralizer of x_ζ in the general linear group of $V(\zeta, x)$. Hence $C(\zeta, x)$ is a classical group by the first part of the proof (cases $j = 2, 3$). Finally let $\operatorname{Re} \zeta = 0$ and $\zeta \neq 0$. If $j \in \{6, 9\}$ then we view $V(\zeta, x)$ as a vector space over the field $\mathbb{R}[x_\zeta]$, which is isomorphic to $\mathbb{C}$. Then $C(\zeta, x)$ is the unitary group defined by the space $V(\zeta, x)$ and the non-degenerate hermitian form whose real (resp. imaginary) part is the restriction of f when $j = 6$ (resp. 9). If $j = 7$ or 10 then there exists an orthogonal basis $e_0, e_1, \dots, e_m$ of $V(\zeta, x)$ such that $f(e_k, e_k) = 1$ ($j=7$) or $f(e_k, e_k) = \pm 1$ ($j=10$) and $x(e_k) = e_k \zeta$ for all k . Let W be the complex linear span of $e_0, e_1, \dots, e_m$. Then $C(\zeta, x)$ is isomorphic to the unitary group defined by the space W and the restriction of f to W , which is a non-degenerate hermitian or skew-hermitian form.

Hence in all cases $C(\zeta, x)$ is a classical group and the proof is completed.

By inspection of the proof we find that k is given by the following table (at present ignore the third column).

TABLE I

j	ζ	λ	k
1	all ζ	all λ	1
2,3	$\zeta = \bar{\zeta}$	$\lambda = \bar{\lambda}$	$k = j$
	$\zeta \neq \bar{\zeta}$	$\lambda \neq \bar{\lambda}$	1
4	$\zeta = -\bar{\zeta}$	$ \lambda = 1$	4
	$\zeta \neq -\bar{\zeta}$	$ \lambda \neq 1$	1
$5 \leq j \leq 10$	$\zeta = 0$	$\lambda = \pm 1$	$k = j$
5,8	$\zeta \neq 0$	$\lambda \neq \pm 1$	1
6,7,9,10	$\zeta \neq \pm \bar{\zeta}$	$\lambda \neq \bar{\lambda} \neq \lambda^{-1}$	1
	$\zeta = -\bar{\zeta} \neq 0$	$\lambda = \bar{\lambda}^{-1} \neq \pm 1$	4
6,9	$\zeta = \bar{\zeta} \neq 0$	$\lambda = \bar{\lambda} \neq \pm 1$	2
7,10	$\zeta = \bar{\zeta} \neq 0$	$\lambda = \bar{\lambda} \neq \pm 1$	3

5. REDUCTION TO THE NILPOTENT CASE

In this section we show that we can determine the closure of an arbitrary orbit O of a classical group G in its Lie algebra L , provided that we know how to find the closures of nilpotent orbits (for all classical groups).

Every $x \in L$ has a unique Jordan decomposition $x = x_s + x_n$, where x_s is semisimple, x_n is nilpotent, both x_s and x_n are in L , and $[x_s, x_n] = 0$. We refer to x_s (resp. x_n) as the semisimple (resp. nilpotent) component of x .

THEOREM 2. *Let G and L be as above and $x, y \in L$. Then $y \in \overline{G \cdot x}$ holds iff $a \cdot y = x_s + z$ for some $a \in G$ and some*

$$z \in \overline{C_G(x_s) \cdot x_n}.$$

Proof. The sufficiency is obvious. Let us assume that $y \in \overline{G \cdot x}$. Then there exists a sequence (a_k) in G such that $\lim(a_k \cdot x) = y$, as $k \rightarrow \infty$. The map which associates to each $u \in L$ its semisimple component

u_s (in the Jordan decomposition $u = u_s + u_n$) is not continuous. But all elements $a_k \cdot x$, considered as linear operators in V , are similar to each other. Therefore we are able to use a result of Parthasarathy [8, Corollary 3.6] and conclude that $\lim(a_k \cdot x_s) = y_s$, as $k \rightarrow \infty$. (Note that $a_k \cdot x_s$ is the semisimple component of $a_k \cdot x$.) Since G is a real algebraic group and x_s is semisimple, the orbit $G \cdot x_s$ is closed in L , [1, Proposition 10.1]. It follows that $y_s \in G \cdot x_s$, and so there exists $a \in G$ such that $a \cdot y_s = x_s$. By [9, Theorem 2.9.7] the orbit $G \cdot x_s$ is a regular submanifold of L . Thus if U is a neighborhood of 1 in G then $U \cdot y_s$ is a neighbourhood of y_s in $G \cdot x_s$. It follows that there exists a sequence (b_k) in G such that $\lim b_k = 1$, as $k \rightarrow \infty$, and $b_k a_k \cdot x_s = y_s$. Now we have

$$\begin{aligned} \lim(ab_k a_k \cdot x_n) &= \lim(ab_k a_k \cdot x) - \lim(ab_k a_k \cdot x_s) \\ &= \lim(ab_k) \cdot \lim(a_k \cdot x) - a \cdot y_s \\ &= a \cdot y - x_s. \end{aligned}$$

Set $z = a \cdot y - x_s$. Note that $ab_k a_k \cdot x_s = a \cdot y_s = x_s$, and so $ab_k a_k \in C_G(x_s)$. Since $z = \lim(ab_k a_k \cdot x_n)$, we indeed have $z \in \overline{C_G(x_s) \cdot x_n}$ and the proof is completed.

REMARK. More generally, let H be an affine real algebraic group and H^0 the identity component of H when H is viewed as a Lie group. Then Theorem 2 is valid for any subgroup G of H containing H^0 , and its Lie algebra L .

Let O_1 and O_2 be the G -orbits in L such that $O_1 \leq O_2$. By Theorem 2 we can choose $x \in O_2$ and $y \in O_1$ such that $y = x_s + z$ where $z = y_n$ satisfies

$$z \in \overline{C_G(x_s) \cdot x_n}. \quad (5.1)$$

By Theorem 1 we have

$$C_G(x_s) = G^{(1)} \times \dots \times G^{(m)} \quad (5.2)$$

where $G^{(k)}$ are classical groups. Then

$$Z_L(x_s) = L^{(1)} \oplus \dots \oplus L^{(m)} \quad (5.3)$$

where $L^{(k)}$ is the Lie algebra of $G^{(k)}$. Since x_n and z are in $Z_L(x_s)$, we have decompositions

$$\begin{aligned}x_n &= x_n^{(1)} + \dots + x_n^{(m)}, \quad x_n^{(k)} \in L^{(k)}; \\z &= z^{(1)} + \dots + z^{(m)}, \quad z^{(k)} \in L^{(k)}.\end{aligned}$$

It follows from (5.1) - (5.3) that

$$z^{(k)} \in \overline{G^{(k)} \cdot x_n^{(k)}}, \quad 1 \leq k \leq m.$$

Consequently, the problem of describing the closure of an arbitrary G -orbit in L reduces to the problem of describing the closure of nilpotent orbits for all classical groups.

6. CONJUGACY CLASSES

Section 3-5 dealt with the adjoint action of G on L but similar results are valid for the action of G on itself via inner automorphisms. For $x, y \in G$ we shall write $x \cdot y = xyx^{-1}$. Then the G -orbit $G \cdot y = \{xyx^{-1} : x \in G\}$ is the conjugacy class of G containing y .

Each $x \in G$ has a unique (multiplicative) Jordan decomposition $x = x_s x_u$, where x_s is semisimple, x_u is unipotent, both x_s and x_u are in G , and $x_s x_u = x_u x_s$. We refer to x_s (resp. x_u) as the semisimple (resp. unipotent) component of x .

For $x \in G$ and $\lambda \in \mathbb{C}^*$ let

$$V(\lambda, x) = \text{Ker}(x - \lambda)^{\infty}$$

if $F = \mathbb{C}$, and otherwise

$$V(\lambda, x) = \text{Ker}(x^2 - 2 \operatorname{Re}(\lambda)x + |\lambda|^2)^{\infty}.$$

If $j \in \{5, 8\}$ and $\lambda \neq \pm 1$ then $V(\lambda, x)$ is totally isotropic. The same is true if $j \in \{4, 6, 7, 9, 10\}$ and $|\lambda| \neq 1$. We now define the subspaces $V'(\lambda, x)$ by

$$V'(\lambda, x) = \begin{cases} V(\lambda, x) & \text{for } j \in \{1, 2, 3\}, \\ V(\lambda, x) + V(\lambda^{-1}, x) & \text{for } j \in \{5, 8\}, \\ V(\lambda, x) + V(\overline{\lambda}^{-1}, x) & \text{for } j \in \{4, 6, 7, 9, 10\}. \end{cases}$$

If $4 \leq j \leq 10$ then the subspaces $V'(\lambda, x)$ are non-degenerate and orthogonal to each other. Hence we obtain the primary decomposition with respect to x :

$$V = \sum_{\lambda \in \Omega_j^*} V'(\lambda, x) \quad (6.1)$$

where

$$\begin{aligned} \Omega_1^* &= \mathbb{C}^*, \\ \Omega_2^* &= \Omega_3^* = \{\lambda \in \mathbb{C}^*: \operatorname{Im} \lambda \geq 0\}, \\ \Omega_4^* &= \{\lambda \in \mathbb{C}^*: |\lambda| \geq 1\}, \\ \Omega_5^* &= \Omega_8^* = \{\lambda \in \mathbb{C}^*: \operatorname{Im} \lambda > 0\} \cup \{\lambda \in \mathbb{R}^*: |\lambda| \geq 1\}, \\ \Omega_6^* &= \Omega_7^* = \Omega_9^* = \Omega_{10}^* = \{\lambda \in \mathbb{C}^*: |\lambda| \geq 1, \operatorname{Im} \lambda \geq 0\}. \end{aligned}$$

Now one can define for a semisimple element $x \in L$ the elements x_λ , x'_λ , and the groups $G(\lambda, x)$ and $C(\lambda, x)$ similarly to the definition of x_ζ , x'_ζ , $G(\zeta, x)$ and $C(\zeta, x)$ in section 4.

Then we have

$$C_G(x) = \prod_{\lambda \in \Omega_j^*} C(\lambda, x) \quad (6.2)$$

and the analog of Theorem 1 is valid. Thus each $C(\lambda, x)$ is a classical group of k -th series, where k depends only on j and λ . The values of k are given in Table I.

The analog of Theorem 2 is also valid.

THEOREM 2'. *Let $x, y \in G$ where G is a classical group. Then $y \in \overline{G \cdot x}$ holds iff $aya^{-1} = x_g z$ for some $a \in G$ and some*

$$z \in \overline{C_G(x_g) \cdot x_u}.$$

In view of Theorem 2' the problem of describing the closure of an arbitrary conjugacy class of G reduces to the problem of describing the closure of unipotent conjugacy classes (in all classical groups). The latter problem reduces via the exponential map to the problem of describing the closure of nilpotent orbits in Lie algebras of classical groups. Indeed we have the following well-known lemma.

LEMMA 3. *Let G be a classical group and L its Lie algebra. Then the restriction of the exponential map $L \rightarrow G$ gives a G -equivariant homeomorphism of the set of nilpotent elements of L with the set of unipotent elements of G .*

7. THE CASES $j = 1, 2, 3, 5, 8$

In these cases the answer to our problem (for nilpotent orbits) is given by the following theorem.

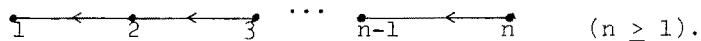
THEOREM 4. *Let G belong to the j -th series where $j \in \{1, 2, 3, 5, 8\}$. If $x, y \in L$ are nilpotent elements then $G \cdot x \leq G \cdot y$ holds iff $\text{rank}(x^k) \leq \text{rank}(y^k)$ for all $k (\geq 0)$.*

For $j \in \{1, 5, 8\}$ this result is due to M. Gerstenhaber [5], [6]. In fact his results are more general since he works over an arbitrary algebraically closed field (in cases $j = 5$ or 8 the characteristic is $\neq 2$). The cases $j = 2, 3$ can be treated in the same way as the case $j = 1$. The case $j = 1$ was also dealt with by J. Dixmier [3], and the cases $j = 5, 8$ by W. Hesselink [7] (in a more general setting) and by the author [4].

8. CHROMOSOMES

In order to state our results in the remaining cases we need certain combinatorial gadgets which we call chromosomes. Roughly speaking they are Young diagrams some of whose rows are filled with alternating + and - signs. A precise definition is given below.

A string (non-polarized) S is a directed graph isomorphic to



The vertex 1 is the head and the vertex n the tail of this string; n is its rank. If we assign alternating + and - signs to the vertices of this string we obtain a polarized string. A polarized string is positive or negative depending on whether its tail carries + or - sign. The signature of a polarized string S is the ordered pair

$$\text{sign } S = (r^+, r^-) \quad (8.1)$$

where r^+ and r^- are the numbers of vertices of S with label + and -, respectively. If S is non-polarized we define its signature by (8.1) where now $r^+ = r^- = \frac{1}{2}n$, n being the rank of S .

A gene is an isomorphism class of strings; it may be polarized or non-polarized. There are precisely three genes of rank n for each positive integer n : the non-polarized gene of rank n , the positive gene of rank n and the negative gene of rank n . We shall denote these three genes in that order by $g(n)$, $g^+(n)$, and $g^-(n)$.

Let Σ be the free abelian group having the set of genes as its basis; the elements of Σ will be written as finite formal integral linear combinations of genes. A chromosome is an element of Σ which is a non-negative integral linear combination of genes. The set of all chromosomes will be denoted by Σ^+ ; it is clearly a free commutative monoid on the set of genes. If $X = Y + Z$ where Y and Z are chromosomes then we say that X contains Y , and we write $X \supset Y$ or $Y \subset X$. If g is a gene and m (> 0) the integer such that $X \supset mg$ and $X \not\supset (m+1)g$ then we say that m is the multiplicity of g in X . The signature of a chromosome X is by definition the sum of the signatures of constituent genes (taking into account the multiplicities). Similarly one defines the rank of X .

A chromosome is called even, (odd, polarized, or non-polarized) if all its constituent genes have even rank (odd rank, are polarized, or are non-polarized), respectively. Every chromosome X admits a unique decomposition $X = e(X) + o(X)$ where $e(X)$ is an even and $o(X)$ an odd chromosome.

A variety is a submonoid of Σ^+ . By Π resp. Λ we denote the varieties of polarized resp. non-polarized chromosomes. If Φ is a variety, so is

$$k\Phi = \{kX: X \in \Phi\},$$

for every non-negative integer k . If Φ is a variety and n (≥ 0) an integer then $\Phi(n)$ denotes the set of all $X \in \Phi$ having rank n . If Φ and Ψ are varieties then (Φ, Ψ) denotes the variety consisting of all chromosomes X such that $e(X) \in \Phi$ and $o(X) \in \Psi$.

We shall need in the sequel the following five varieties

$$\Pi, (\Lambda, \Pi), (\Pi, \Lambda), (2\Lambda, \Pi), (\Pi, 2\Lambda). \quad (8.2)$$

If g is a gene of rank n (≥ 1) we define g' as follows

$$\begin{aligned} g' &= g(n-1) \quad \text{if } g = g(n), \\ &= g^+(n-1) \quad \text{if } g = g^+(n), \\ &= g^-(n-1) \quad \text{if } g = g^-(n). \end{aligned}$$

If g is a gene of rank 1 we have $g' = 0$. If X is a chromosome then we define X' by applying this "prime" operation to each of its genes. Now we define inductively the chromosomes $X^{(k)}$, $k \geq 0$, as follows.

$$X^{(0)} = X, \quad X^{(k+1)} = (X^{(k)})' \quad \text{for } k \geq 0.$$

For instance, if

$$X = 2g(3) + g^-(2) + g(2) + g^+(4)$$

then

$$\begin{aligned} X' &= X^{(1)} = 2g(2) + g^-(1) + g(1) + g^+(3), \\ X^{(2)} &= 2g(1) + g^+(2), \quad X^{(3)} = g^+(1) \\ X^{(k)} &= 0 \quad \text{for } k \geq 4. \end{aligned}$$

For ordered pairs of real numbers the inequality $(p,q) \leq (r,s)$ stands for $p \leq r$ and $q \leq s$. We say that a chromosome X dominates a chromosome Y , and we write $X \geq Y$, if

$$\text{sign}(X^{(k)}) \geq \text{sign}(Y^{(k)}), \quad \forall k.$$

When restricted to one of the varieties (8.2) the dominance relation is a partial order.

9. DESCRIPTION OF NILPOTENT ORBITS

Let G be a classical group belonging to the j -th series and let L be its Lie algebra. In view of Theorem 4 we restrict ourselves to the cases $j = 4, 6, 7, 9, 10$. Let O be an orbit of G in L and $x \in O$. Recall that G is viewed as the group of automorphism of an F -vector space V preserving a non-degenerate form f on V . We shall make use of several well-known facts for which we refer the reader to [2]. There exists a direct decomposition

$$V = V_1 \oplus \dots \oplus V_r$$

into x -invariant non-zero non-degenerate subspaces which are orthogonal to each other and moreover the summands $V_1, \dots, V_r$ cannot be further decomposed in the same sense. Let x_k and f_k be the restrictions of x

and f , respectively, to V_k . Let G_k be the classical group (belonging to the same series) which is defined by the pair (V_k, f_k) . Finally let L_k be the Lie algebra of G_k and let O_k be the G_k -orbit in L_k containing the element x_k .

Let Δ be the type corresponding to the orbit O and Δ_k type corresponding to O_k ; see [2] for the definition of types. Then we have

$$\Delta = \Delta_1 + \dots + \Delta_r.$$

Thus each type is a sum of indecomposable types and as shown in [2] this decomposition is unique.

The nilpotent indecomposable types are denoted by:

$$\begin{aligned} &\Delta_m^\epsilon(0) \text{ if } j = 4; \\ &\Delta_m^\epsilon(0), m \text{ even}; \Delta_m(0,0), m \text{ odd if } j = 6; \\ &\Delta_m(0,0), m \text{ even}; \Delta_m^\epsilon(0,0), m \text{ odd if } j = 7; \\ &\Delta_m(0,0), m \text{ even}; \Delta_m^\epsilon(0), m \text{ odd if } j = 9; \\ &\Delta_m^\epsilon(0,0), m \text{ even}; \Delta_m(0,0), m \text{ odd if } j = 10. \end{aligned}$$

In this list $\epsilon = \pm$ and $m (\geq 0)$ is an integer.

We have to change the definition of Δ_m^+ and Δ_m^- which is used in [2]. Since this is an important point we shall give representative (V, f, x) , $x \in L$ for each of these types.

If $j = 4$ a representative (V, f, x) of $\Delta_m^\epsilon(0)$ is given by:

$$\begin{aligned} &V \text{ a complex vector space} \\ &e_0, e_1, \dots, e_m \text{ a basis of } V, \\ &x(e_k) = e_{k+1}i \quad (i = \sqrt{-1}, 0 \leq k < m), \quad x(e_m) = 0, \\ &f \text{ a hermitian form on } V, \\ &f(e_r, e_s) = \epsilon \delta_{r+s, m} \quad (\delta_{ij} \text{ Kronecker symbol}). \end{aligned} \tag{9.1}$$

If $j = 6$ a representative (V, f, x) of $\Delta_m^\epsilon(0)$, m even, $m = 2m'$, is given by:

$$\begin{aligned} &V \text{ is a real vector space,} \\ &e_0, e_1, \dots, e_m \text{ a basis of } V, \\ &\left. \begin{aligned} x(e_k) &= e_{k+1} \quad (0 \leq k < m'), \\ x(e_k) &= -e_{k+1} \quad (m' \leq k < m), \quad x(e_m) = 0, \end{aligned} \right\} \\ &f \text{ a symmetric bilinear form on } V, \\ &f(e_r, e_s) = \epsilon \delta_{r+s, m}. \end{aligned} \tag{9.2}$$

If $j = 7$ a representative (V, f, x) of $\Delta_m^{\varepsilon}(0, 0)$, m odd, is given by:

V a (right) quaternionic vector space,
 $e_0, e_1, \dots, e_m$ a basis of V ,
 x given by (9.1),
 f a skew-hermitian form on V ,
 $f(e_r, e_s) = -\varepsilon \delta_{r+s, m}$.

If $j = 9$ a representative (V, f, x) of $\Delta_m^{\varepsilon}(0, 0)$, m odd, $m = 2m' - 1$ is given by:

V a real vector space,
 $e_0, e_1, \dots, e_m$ a basis of V ,
 x given by (9.2),
 f a skew-symmetric form on V ,
 $f(e_r, e_s) = \varepsilon \delta_{r+s, m}$ for $r < s$.

If $j = 10$ then a representative (V, f, x) of $\Delta_m^{\varepsilon}(0, 0)$, m even, is given by:

V a (right) quaternionic vector space,
 $e_0, e_1, \dots, e_m$ a basis of V ,
 x given by (9.1),
 f a hermitian form on V ,
 $f(e_r, e_s) = \varepsilon \delta_{r+s, m}$.

The nilpotent types Δ_m^+ and Δ_m^- can be distinguished by a simple inequality. Namely, if (V, f, x) is one of the representatives listed above then for all $v \in V$ we have

$$\begin{aligned} (-i)^m \varepsilon f(v, x^m(v)) &\geq 0 \text{ if } j = 4, \\ i^m \varepsilon f(v, x^m(v)) &\geq 0 \text{ if } j = 6, 10, \\ i^{m-1} \varepsilon f(v, x^m(v)) &\geq 0 \text{ if } j = 7, 9. \end{aligned}$$

If $j = 6$ (resp. 9) a representative (V, f, x) of $\Delta_m(0, 0)$ with m odd (resp. even) is given by:

V a real vector space,
 $e_0, \dots, e_m, e'_0, \dots, e'_m$ a basis of V ,
 $x(e_k) = e_{k+1}$ ($0 \leq k < m$), $x(e_m) = 0$,
 $x(e'_k) = -e'_{k+1}$ ($0 \leq k < m$), $x(e'_m) = 0$,
 f is a symmetric (resp. skew-symmetric) form on V ,
 $f(e_r, e_s) = f(e'_r, e'_s) = 0$, $f(e_r, e'_s) = \delta_{r+s, m}$.

If $j = 7$ then a representative of $\Delta_m(0,0)$ with m even has the same description as the representative of $\Delta_m^\varepsilon(0,0)$ with m odd. (The choice of $\varepsilon = \pm$ is not important since both represent the same type.) Similarly if $j = 10$ then a representative of $\Delta_m(0,0)$ with m odd has the same description as the representative of $\Delta_m^\varepsilon(0,0)$ with m even.

10. NILPOTENT ORBITS AND CHROMOSOMES

To each nilpotent indecomposable type Δ of the j -th series we associate a chromosome X belonging to a variety Φ_j ($j = 4, 6, 7, 9, 10$).

TABLE II

j	Δ	X	Φ_j
4	$\Delta_m^\varepsilon(0)$	$g^\varepsilon(m+1)$	Π
6	$\Delta_m^\varepsilon(0)$, m even	$g^\varepsilon(m+1)$	$(2\Lambda, \Pi)$
	$\Delta_m(0,0)$, m odd	$2g(m+1)$	
7	$\Delta_m(0,0)$, m even	$g(m+1)$	(Π, Λ)
	$\Delta_m^\varepsilon(0,0)$, m odd	$g^\varepsilon(m+1)$	
9	$\Delta_m(0,0)$, m even	$2g(m+1)$	$(\Pi, 2\Lambda)$
	$\Delta_m^\varepsilon(0)$, m odd	$g^\varepsilon(m+1)$	
10	$\Delta_m^\varepsilon(0,0)$, m even	$g^\varepsilon(m+1)$	(Λ, Π)
	$\Delta_m(0,0)$, m odd	$g(m+1)$	

Now let G be a classical group and L its Lie algebra. Let V , F , and f be as in section 1, and write $n = \dim_F V$. If $j = 4, 6, 10$ let n^+ (resp. n^-) be the maximum F -dimension of a positive (resp. negative) definite subspace of V . Then we set

$$\text{sign } f = (n^+, n^-). \quad (10.1)$$

If $j = 7$ or 9 we define $\text{sign } f$ by (10.1) where now $n^+ = n^- = \frac{1}{2}n$.

The following lemma is an immediate consequence of our definitions and the description of G -orbits in L given in the previous section.

LEMMA 5. By associating to each nilpotent indecomposable type Δ of G the chromosome $X \in \Phi_j$, given by Table II, we obtain a bijection from the set of nilpotent orbits of G in L to the set of chromosomes $X \in \Phi_j$ satisfying the condition $\text{sign } X = \text{sign } f$.

For instance let $j = 10$ and $\text{sign } f = (2, 2)$. Then there are six nilpotent orbits in L :

$$\begin{aligned} & 2\Delta_0^+(0,0) + 2\Delta_0^-(0,0), \\ & \Delta_0^+(0,0) + \Delta_0^-(0,0) + \Delta_1(0,0), \\ & 2\Delta_1(0,0), \\ & \Delta_0^+(0,0) + \Delta_2^-(0,0), \\ & \Delta_0^-(0,0) + \Delta_2^+(0,0), \\ & \Delta_3(0,0). \end{aligned}$$

These are also listed in [2, p.351] but the superscripts $\pm$ do not coincide due to different definition of these types. The chromosomes in (Λ, Π) corresponding to these types are: $2g^+(1) + 2g^-(1)$, $g^+(1) + g^-(1) + g(2)$, $2g(2)$, $g^+(1) + g^-(3)$, $g^-(1) + g^+(3)$, and $g(4)$, respectively.

11. MAIN RESULT

Let G be a classical group of the j -th series with $j \in \{4, 6, 7, 9, 10\}$. For a nilpotent orbit O of G in L let $X(O)$ be the chromosome associated to O as in the previous section.

Then we have the following result.

THEOREM 6. If O_1 and O_2 are nilpotent orbits of G in L then $O_1 \leq O_2$ holds iff $X(O_1) \leq X(O_2)$.

Proof of necessity. Let V and f be the vector space and the form which define G . Let $x \in L$ be a nilpotent element. For integers k (≥ 0) we define forms $f_{x,k}$ on V : if $j = 4$ then

$$f_{x,k}(v_1, v_2) = (-1)^k f(v_1, x^k(v_2)),$$

and if $j \in \{6, 7, 9, 10\}$ then

$$f_{x,k}(v_1, v_2) = i^{k'} f(v_1, x^k(v_2))$$

where $k' = k$ for k even and $k' = k - 1$ for k odd. For $j = 4$ the forms $f_{x,k}$ are hermitian; for $j = 6$ (resp. 9) they are symmetric for k even (resp. odd) and skew-symmetric for k odd (resp. even); for $j = 7$ (resp. 10) they are hermitian for k odd (resp. even) and skew-hermitian for k even (resp. odd). Since these forms are in general degenerate, we have to extend our definition of signature to such forms g . We set $\text{sign}(g) = \text{sign}(\tilde{g})$ where $\tilde{g}$ is the (nondegenerate) form induced by g on $V/\text{Rad } g$.

If (g_k) is a sequence of such forms (all of the same type) and $g = \lim_{k \rightarrow \infty} g_k$ then it is clear that

$$\text{sign}(g) \leq \lim_{k \rightarrow \infty} \text{sign}(g_k), \quad (11.1)$$

i.e., if $\text{sign}(g) = (n_+, n_-)$ and $\text{sign}(g_k) = (n_+^{(k)}, n_-^{(k)})$ then $n_+^{(k)}$ and $n_-^{(k)}$ are constant for large k and $n_+ \leq n_+^{(k)}$, $n_- \leq n_-^{(k)}$ hold for large k .

Writing $O = G \cdot x$ we claim that

$$\text{sign}(f_{x,k}) = \text{sign } X(O)^{(k)} \quad (11.2)$$

holds for all k . It suffices to prove this equality in the case when the type Δ determined by O is indecomposable. Then (11.2) follows immediately from the description of representatives of such types given in section 9.

Now let $x_1 \in O_1$ and $x_2 \in O_2$. Since $O_1 \leq O_2$ by hypothesis, we can choose a sequence (y_m) in O_2 such that $x_1 = \lim_{m \rightarrow \infty} y_m$ as $m \rightarrow \infty$. Then for each k we have

$$\lim_{m \rightarrow \infty} f_{y_m,k} = f_{x_1,k}.$$

Since $\text{sign}(f_{y_m,k}) = \text{sign}(f_{x_2,k})$ for all m , it follows from (11.1) that

$$\text{sign}(f_{x_1,k}) \leq \text{sign}(f_{x_2,k})$$

for all k . In view of (11.2) this gives

$$\text{sign } X(O_1)^{(k)} \leq \text{sign } X(O_2)^{(k)}$$

for all k . This means that $X(O_1) \leq X(O_2)$, which is the condition stated in the theorem.

Sufficiency (outline of the proof). The chromosomes $X = X(O_1)$ and $Y = X(O_2)$ satisfy $X \leq Y$, by hypothesis, and also $\text{sign } X = \text{sign } Y = \text{sign } f$. Without any loss of generality we may also assume that $X \neq Y$ and that $X \leq Z \leq Y$ for $Z \in \Phi_j$ implies $Z = X$ or $Z = Y$. The rest of the proof is divided into two parts. The first part is of combinatorial nature: we describe all pairs of chromosomes (X, Y) in Φ_j satisfying the conditions that we just mentioned. (See the next section for the precise statement of these results.) It turns out that if we delete the common genes of X and Y (taking into account the multiplicities of course) then at most four genes will be left in each X and Y . In the second part of the proof for each pair of chromosomes (X, Y) satisfying the above conditions and moreover such that X and Y are disjoint we construct a continuous 1-parameter family $x(t) \in L$ ($t \in \mathbb{R}$) such that $x(t) \in O_2$ for $t > 0$ and $x(0) \in O_1$.

For instance when $j = 4$ we may assume that (X, Y) is one of the following pairs

$$(g^{\epsilon_1(m+1)+g^{\epsilon_2(n+1)}, g^{\epsilon_2(m)+g^{\epsilon_1(n+2)}, n-m = 2k+1 \quad (11.3)$$

$$(g^{\epsilon(m+1)+g^{\epsilon(n+1)}, g^{\epsilon(m-1)+g^{\epsilon(n+3)}), 1 \leq m \leq n \quad (11.4)$$

$$(g^{-\epsilon_1(m+1)+g^{\epsilon_1(n+1)}, g^{-\epsilon_2(m)+g^{\epsilon_2(n+2)}), n-m = 2k \quad (11.5)$$

where $k (\geq 0)$ is an integer and $\epsilon, \epsilon_1, \epsilon_2$ are arbitrary signs $\pm$. In order to show that $O_1 \leq O_2$ when (X, Y) is given by (11.3) we construct the following family $x(t) \in L$. First we choose a basis $e_0, \dots, e_m, e'_0, \dots, e'_n$ of V such that

$$f(e_r, e'_s) = 0, f(e_r, e_s) = \epsilon_1 \delta_{r+s, m}, f(e'_r, e'_s) = \epsilon_2 \delta_{r+s, n}$$

holds for all relevant r and s . Then the linear operators $x(t): V \rightarrow V$ are defined by specifying that

$$x(t)e_r = ie_{r+1} \quad (0 \leq r < m), x(t)e_m = it e'_{n-k};$$

$$x(t)e'_r = ie'_{r+1} \quad \text{for } r \neq k, n;$$

$$x(t)e'_k = i(e'_{k+1} + \epsilon_1 \epsilon_2 t e_0), x(t)e'_n = 0.$$

It is easy to check that

$$f(x(t)v_1, v_2) + f(v_1, x(t)v_2) = 0$$

for all $v_1, v_2 \in V$, i.e., $x(t) \in L$. An easy computation gives

$$\begin{aligned} x(t)^n e'_0 &= i^n (e'_n + \varepsilon_1 \varepsilon_2 t^2 e'_{n-1}), \quad x(t)^{n+1} e'_0 = \varepsilon_1 \varepsilon_2 t^2 i^{n+1} e'_n, \\ x(t)^{n+2} e'_0 &= 0, \quad x(t)^{m-1} (e_1 - t e'_{k+1}) = i^{m-1} (e_m - t e'_{k+m}), \\ x(t)^m (e_1 - t e'_{k+1}) &= 0. \end{aligned}$$

Thus if $t \neq 0$ then e'_n and $e_m - t e'_{k+m}$ form a basis of $\text{Ker } x(t)$. Writing $e = e_1 - t e'_{k+1}$, we have

$$\varepsilon_1 (-1)^{n+1} f(e'_0, x(t)^{n+1} e'_0) = t^2 = \varepsilon_2 (-1)^{m-1} f(e, x(t)^{m-1} e).$$

Recalling the definition of the types $\Delta_m^\varepsilon(0)$ when $j = 4$ (see section 9), we infer that for $t \neq 0$ we have $x(t) \in O_2$, while on the other hand $x(0) \in O_1$. Therefore $O_1 \leq O_2$.

12. MUTATIONS

For each variety Φ listed in (8.2) we shall define some transformations of chromosomes which we call Φ -mutations. A chromosome $X \in \Phi$ is transformed by a Φ -mutation into another chromosome $Y \in \Phi$ such that $X \leq Y$, $X \neq Y$, and $\text{sign } X = \text{sign } Y$. We shall indicate such mutation symbolically by $X \rightarrow Y$. We warn the reader that we may have two Φ -mutations $X \rightarrow Y$ and $X \rightarrow Z$ with $Y \neq Z$ (see the example below).

If $X \rightarrow Y$ is a Φ -mutation and $Z \in \Phi$ then also $X + Z \rightarrow Y + Z$ is a Φ -mutation. Conversely if $X, Y, Z \in \Phi$ and $X + Z \rightarrow Y + Z$ is a Φ -mutation then also $X \rightarrow Y$ is a Φ -mutation. Hence it suffices to list only the Φ -mutations $X \rightarrow Y$ such that X and Y are disjoint (i.e. they have no common genes). Such Φ -mutations we call primitive.

The primitive Π -mutations are defined by:

$$g^\varepsilon(m) + g^{-\varepsilon}(n) \rightarrow g^{-\varepsilon}(m-1) + g^\varepsilon(n+1), \quad m \leq n; \quad (12.1)$$

$$g^\varepsilon(m) + g^\varepsilon(n) \rightarrow g^\varepsilon(m-2) + g^\varepsilon(n+2), \quad 1 < m \leq n; \quad (12.2)$$

$$g_\varepsilon(m) + g_{-\varepsilon}(n) \rightarrow g_\varepsilon(m-1) + g_{-\varepsilon}(n+1), \quad m \leq n, \quad (12.3)$$

where

$$g_{\varepsilon}(n) = g^{(-1)^{n-1}\varepsilon}(n), \quad \varepsilon = \pm.$$

EXAMPLE. If $X = g^+(5) + g^+(4) + g^-(1)$ and $X \rightarrow Y$ is a Π -mutation then Y is one of the chromosomes $Y_1 = g^-(6) + g^+(4)$, $Y_2 = g^+(5) + g^-(5)$, $Y_3 = g^+(6) + g^+(3) + g^-(1)$, or $Y_4 = g^+(6) + g^+(4)$. The mutations $X \rightarrow Y_1$ and $X \rightarrow Y_2$ are of type (12.1) and the mutations $X \rightarrow Y_3$ and $X \rightarrow Y_4$ are of type (12.3).

If $\Phi = (\Lambda, \Pi)$ or (Π, Λ) then the primitive Φ -mutations are defined by:

$$\begin{aligned} g(m) + g(n) &\rightarrow g^{\varepsilon}(m-1) + g^{-\varepsilon}(n+1), \quad m \leq n; \\ g(m) + g^{\varepsilon}(n) &\rightarrow g^{\varepsilon}(m-1) + g(n+1), \quad m < n; \\ g^{\varepsilon}(m) + g(n) &\rightarrow g(m-1) + g^{\varepsilon}(n+1), \quad m < n; \\ g^{\varepsilon}(m) + g^{-\varepsilon}(n) &\rightarrow g(m-1) + g(n+1), \quad m \leq n; \\ g^{\varepsilon}(m) + g^{\varepsilon}(n) &\rightarrow g^{\varepsilon}(m-2) + g^{\varepsilon}(n+2), \quad 1 < m \leq n; \end{aligned}$$

where $\varepsilon = \pm$.

Finally, if $\Phi = (2\Lambda, \Pi)$ or $(\Pi, 2\Lambda)$ then the primitive Φ -mutations are listed below:

$$\begin{aligned} 2g(m) &\rightarrow g^{\varepsilon}(m-1) + g^{-\varepsilon}(m+1); \\ g^{\varepsilon_1}(m) + g^{\varepsilon_2}(n) &\rightarrow g^{\varepsilon_1}(m-2) + g^{\varepsilon_2}(n+2), \quad 1 < m \leq n; \\ g^{\varepsilon}(m) + g^+(n) + g^-(n) &\rightarrow g^{\varepsilon}(m-2) + 2g(n+1), \quad 1 < m \leq n; \\ g^+(m) + g^-(m) + g^{\varepsilon}(n) &\rightarrow 2g(m-1) + g^{\varepsilon}(n+2), \quad m \leq n; \\ g^+(m) + g^-(m) + g^+(n) + g^-(n) &\rightarrow 2g(m-1) + 2g(n+1), \quad m \leq n; \\ 2g^{\varepsilon}(m) + 2g^{-\varepsilon}(n) &\rightarrow 2g(m-1) + 2g(n+1), \quad m \leq n; \\ g^{\varepsilon}(m) + g^{-\varepsilon}(n) &\rightarrow g^{-\varepsilon}(m-2) + g^{\varepsilon}(n+2), \quad 1 < m \leq n; \\ 2g^{\varepsilon}(m) + g^{-\varepsilon}(n) &\rightarrow 2g(m-1) + g^{\varepsilon}(n+2), \quad m \leq n; \\ g^{\varepsilon}(m) + 2g^{-\varepsilon}(n) &\rightarrow g^{-\varepsilon}(m-2) + 2g(n+1), \quad 1 < m \leq n; \end{aligned}$$

where $\varepsilon = \pm$.

THEOREM 7. *Let Φ be one of the varieties (8.2) and $X, Y \in \Phi$ such that $X \leq Y$ and $\text{sign } X = \text{sign } Y$. Then there exists a finite chain of Φ -mutations*

$$X_0 \rightarrow X_1 \rightarrow \dots \rightarrow X_m \quad (m \geq 0)$$

such that $X_0 = X$ and $X_m = Y$.

The proof of this theorem is similar in nature (but much harder!) to the proof of analogous results for the varieties Λ , $(2\Lambda, \Lambda)$, and $(\Lambda, 2\Lambda)$ which are given in [4] and [7].

We have already indicated how this combinatorial result is used in the proof of our main result (Theorem 6). To be specific, let say $j = 4$. The primitive Π -mutations $X \rightarrow Y$ are defined by (12.1) - (12.3). The pairs (X, Y) which arise in this way are precisely the pairs listed in (11.3) - (11.5). This justifies our claim that in the proof of the main theorem we may assume (X, Y) to be given by one of the pairs (11.3) - (11.5).

13. REMARK ABOUT THE ADJOINT GROUPS

Let G be a classical group, as before, and L its Lie algebra. The corresponding adjoint group is

$$\text{Ad}(G^0) = G^0/Z(G^0)$$

where G^0 is the identity component of G and $Z(G^0)$ the center of G^0 . Clearly $\text{Ad}(G^0)$ acts on L and its orbits in L are the same as G^0 -orbits. On the other hand the G^0 -orbits are precisely the connected components of G -orbits. Since $G = G^0$ if $j \in \{1, 3, 4, 7, 8, 9, 10\}$, it remains to consider G^0 -orbits only in the cases $j = 2, 5, 6$. When $j = 2$ or 5 a G -orbit has one or two components, while for $j = 6$ it may have one, two, or four components. We can also describe the closures of G^0 -orbits in the cases $j = 2, 5, 6$ but the results are too complicated to be stated here.

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